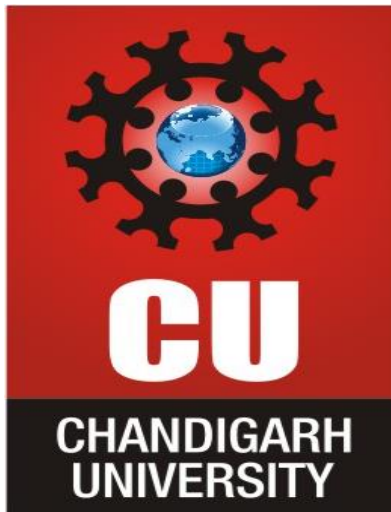




EXPERIMENT : 10

NAME : WASEEM AKRAM

SEMESTER : 5th

SECTION : 24_AIT_KRG-G1-A

UID : 24BAI70006

(A)

Question 1 — Savepoints

Question

Design a transaction on Students_01 where:

- **Insert Alice and keep her.**
- **Insert Bob and create a savepoint.**
- **Insert Charlie, then roll back only Charlie.**
- **Insert Dina.**
- **Roll back everything after sp1.**

Solution:

```
BEGIN TRANSACTION;

INSERT INTO Students_01(ID, name, age, class)
VALUES (7, 'Alice', 18, 10);

SAVEPOINT sp1;

INSERT INTO Students_01(ID, name, age, class)
VALUES (8, 'Bob', 17, 11);

SAVEPOINT sp2;

INSERT INTO Students_01(ID, name, age, class)
VALUES (9, 'Charlie', 16, 9);

-- Remove only Charlie
ROLLBACK TO SAVEPOINT sp2;

-- Continue transaction

INSERT INTO Students_01(ID, name, age, class)
VALUES (10, 'Dina', 15, 8);

-- Remove everything after sp1
ROLLBACK TO SAVEPOINT sp1;

-- Finally undo the entire transaction
ROLLBACK;
```

(B)

Question 2 — Hard Question: Savepoints + Error Handling

Question

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction.

If any insert fails due to invalid data, only that particular insert should be rolled back while preserving the previous successful inserts using savepoints.

The system should also display messages for successful and failed insertions.

Solution:

```
DO $$
```

```
BEGIN
```

```
-- Student 1
```

```
BEGIN
```

```
    INSERT INTO students(name, age, class)
```

```
    VALUES ('Anisha', 16, 8);
```

```
    RAISE NOTICE 'Anisha inserted successfully';
```

```
EXCEPTION WHEN OTHERS THEN
```

```
    RAISE NOTICE 'Anisha insertion failed';
```

```
END;
```

```
-- Student 2
```

```
BEGIN
```

```
    INSERT INTO students(name, age, class)
```

```
    VALUES ('Mayank', 19, 9);
```

```
    RAISE NOTICE 'Mayank inserted successfully';
```

```
EXCEPTION WHEN OTHERS THEN
```

```
    RAISE NOTICE 'Mayank insertion failed';
```

```
END;
```

```
-- Student 3
```

```
BEGIN
```

```
    INSERT INTO students(name, age, class)
```

```
    VALUES ('Anisha', 17, 8);
```

```
    -- ERROR because Anisha already exists
```

```
        RAISE NOTICE 'Anisha inserted successfully';

EXCEPTION WHEN OTHERS THEN
    RAISE NOTICE 'Anisha insertion failed - duplicate name';
END;


-- Student 4
BEGIN
    INSERT INTO students(name, age, class)
    VALUES ('Riya', 18, 10);

    RAISE NOTICE 'Riya inserted successfully';

EXCEPTION WHEN OTHERS THEN
    RAISE NOTICE 'Riya insertion failed';
END;

END;
$$;

SELECT * FROM students;;
```